

TABLE II
Insulation Resistance Testers from Different Manufacturers

Specifications		Fluke Fluke 1507	IDEAL Model: 61-797	HIOKI IR4056-20	Metrix + DIT 3125
Insulation specifications	Measurement range	1507: 0.01M Ω -10G Ω	Up to 20G Ω	Up to 4000M Ω	0-1000G Ω
	Test voltage	50V, 100V, 250V, 500V, 1000V	50V, 100V, 250V, 500V, 1000V	50V, 100V, 250V, 500V, 1000V	30-600V
	Test voltage accuracy	+20%, -0%	0-20%	2% $\pm$ 5 digits	2% $\pm$ 3 digits
	Short circuit current (mA)	1mA nominal	1mA nominal	1.2mA or less	1.3mA
	Auto discharge	Discharge time <0.5 seconds for C = 1 μ F or less	Discharge time <1 second for C <1 μ F	Yes	Yes
	Maximum capacitive load	Operable with up to 1 μ F load	Operable with up to 1 μ F load	Not specified	AC 1200V/10 sec
	Measure accuracy	$\pm$ 3% +5 (below 100V)	—	4%	>100G Ω $\pm$ 20%
Earth-bond resistance measurement	Range	20 Ω /200 Ω /2000 Ω /20k Ω		10-1000 Ω	
	Resolution	0.01 Ω /0.1 Ω /1.0 Ω /0.01k Ω	0.01 ohm to 40.00 k Ω	0.01 Ω (10 Ω range)	
	Accuracy	$\pm$ (1.5% +3)		3% $\pm$ 2 digits	
	Overload protection	2V RMS or DC	600V AC RMS/DC	600V AC	
	Open circuit test voltage	>4.0V, <8V	>4.0V, <8V	4.0V to 6.9V	
	Short circuit current	>200.0mA	>200.0mA	200mA or more	
Safety		Complies with ANSI/ISA 82.02.01 (61010-1) 2004, CAN/CSA-C22.2 NO. 61010-1-04, and IEC/EN 61010-1 second edition for measurement category IV 600V (CAT IV)	Complies with UL/IEC/EN 61010-1, 61010-031, EN61557, EN 61326-1 + 1A (EMC) Cat IV-600V	CAT III 600V	IEC 61010-1
IP rating		IP40	—		IEC 60529 (IP40)
Dimensions		203 $\times$ 100 $\times$ 50mm	207 $\times$ 95 $\times$ 52mm	159 $\times$ 177 $\times$ 53mm	153 $\times$ 213 $\times$ 95 mm
Weight (gm)		550	630	600	1030

use of grounding system is protection against direct lightning strokes.

It is estimated that at least 15 per cent of the power quality problems are related to improper grounding. Therefore good grounding ensures that these problems do not escalate. Fig. 3 shows an earth ground tester and table III shows the testers from different manufacturers.



Fig. 3: Earth ground tester

Laser distance meter

Laser distance meter can be a very useful tool for measuring distances with good precision without physical contact. In fact it allows for the most sensitive and precise length measurements, faster recordings encompassing large ranges. All these qualities do not come with a single technique so different techniques are used based on specific requirements. Some of the most important techniques used for laser distance meters are as follows:

1. *Triangulation* is a geometric method, useful for distances in the range of a millimetre to many kilometres.

2. *Time-of-flight measurements* (or pulse measurements) are based on measuring the time-of-flight of a

laser pulse from the measurement device to some target and back again. Such methods are typically used for large distances.

3. *The phase-shift method* uses an intensity-modulated laser beam. Compared with other techniques, its accuracy is lower, but it allows unambiguous measurements over larger distances and is more suitable for targets with diffuse reflection.

4. *Frequency modulation methods* involve frequency-modulated laser beams, for example, with a repetitive linear frequency ramp. The distance to be measured can be translated into a



Fig. 4: Distance meter